THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 7: **Work, Energy and Simple Machines** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A mass of 2 kg is raised by 3 m with an acceleration of 2 m/s squared from rest (g = 10). The work done by the lifting force is: 1

(a) 60 J **(b)** 36 J **(c)** 12 J **(d)** 72 J

2. The cost of running a 1.5 kW AC for 8 hours a day for 30 days at 7 rupees per unit is: 1

(a) 2520 rupees **(b)** 25200 rupees **(c)** 840 rupees **(d)** 84 rupees

3. Assertion (A): A fixed pulley has mechanical advantage equal to 1. Reason (R): It only changes the direction of the force. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. Two bodies of masses m and 4m have equal KE. The ratio of their speeds is: 1

(a) 1 : 2 **(b)** 2 : 1 **(c)** 1 : 4 **(d)** 4 : 1

5. A motor of 10 kW lifts a mass of 500 kg at a constant speed. The speed is (g = 10): 1

(a) 2 m/s **(b)** 5 m/s **(c)** 20 m/s **(d)** 0.5 m/s

6. A pump of efficiency 80 per cent receives an input power of 500 W. The useful power is: 1

(a) 500 W **(b)** 625 W **(c)** 80 W **(d)** 400 W

7. Assertion (A): A machine can have an efficiency greater than 100 per cent. Reason (R): Output work cannot exceed input work. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

8. A force of 10 N acts on a body that moves 4 m in the direction opposite to the force. The work done by the force is: 1

(a) 2.5 J **(b)** 40 J **(c)** 0 **(d)** -40 J

9. **Assertion (A):** When the speed of a body is doubled, its kinetic energy becomes four times. 1
Reason (R): The kinetic energy of a body is directly proportional to its speed.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** A single fixed pulley has a mechanical advantage greater than 1. 1
Reason (R): A fixed pulley changes the direction of the applied force.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A man pulls a rope through 20 m with an effort of 50 N to raise a load of 100 N with an ideal machine. Find the distance moved by the load. 2

12. A 1000 W motor lifts 50 kg to 10 m. Find the time if the efficiency is 50 per cent (g = 10). 2

13. Why does a heavy truck need a long distance to stop? Show with KE = F x d for 10000 kg at 20 m/s and F = 50000 N. 2

14. Explain why power consumption of an electric appliance is written in watts and the energy in kWh on the bill. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The bob of a pendulum of mass 0.2 kg is released from a point 0.5 m above the lowest point. Find the maximum speed and the KE and PE when it is 0.2 m above the lowest point (g = 10). 3

16. A skier of mass 70 kg starts at rest at the top of a 50 m high slope and reaches the bottom with a speed of 25 m/s. Find the energy lost to friction and air (g = 10). 3

17. Explain, with the example of a ball on a hill, stable, unstable and neutral equilibrium in terms of potential energy. 3

OR

A body of mass 10 kg is pushed up a smooth 5 m long incline of height 2 m. Find the force needed, the work and MA (g = 10). 3

18. Explain why the work done in lifting a load through a given height by a lever, a pulley and an inclined plane is at least mgh. 3

OR

A pump raises 5000 kg of water to a tank 12 m high in 10 minutes. Find the power in watts and in hp (1 hp = 746 W) (g = 10). 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A 2 kg ball is dropped from the roof of a building 20 m high. Ignore air resistance and take $g = 10$ m/s^2. 5
(i) Find the potential energy of the ball at the top.
(ii) What is its kinetic energy just before it hits the ground?
(iii) Find its speed just before it hits the ground.
(iv) Find the kinetic energy when it is 5 m above the ground and state what happens to the total mechanical energy.

20. Compare an inclined plane (length 5 m, height 1 m), a lever (arms 1 m and 0.25 m) and a pulley system (MA = 4) in lifting a 400 N load: find the ideal effort in each, the work input and the common work output, and say which has the highest MA. 5

OR

Explain how energy is transformed in a coal power plant from the burning of fuel to electricity at home, with the typical efficiency of each step, and calculate the overall efficiency if boiler 80 per cent, turbine 40 per cent, generator 95 per cent. 5

END OF QUESTION PAPER — Total: 40 Marks